Contents

1 Introduction **1**

2 siunitx for the impatient **1**

3 Using the siunitx package **2**

3.1 Numbers . . . 2
3.1.1 Other Tests . . . 2
3.1.2 Lists and ranges of numbers . . . 2
3.2 Angles . . . 3
3.3 Units . . . 3
3.3.1 unit . . . 3
3.3.2 qty . . . 3
3.3.3 qtylist . . . 3
3.3.4 qtyproduct . . . 3
3.3.5 qtyrange . . . 3
3.4 Complex numbers and quantities . . . 3
3.5 The unit macros . . . 4
3.6 Unit abbreviations . . . 6
3.7 Creating new macros . . . 10
3.7.1 Other Tests . . . 10
3.8 Tabular material . . . 10

4 Package control options **11**

4.2 Printing . . . 11
4.2.1 mode, number-mode, unit-mode . . . 11
4.2.2 reset-text-[family|series|shape] . . . 11
4.2.3 propagate-math-font, reset-math-version . . . 12
4.2.4 text-[family|series]-to-math . . . 12
4.2.5 text-font-command . . . 12
4.2.6 text-[sub|super]script-command . . . 12
4.2.7 color, number-color, unit-color . . . 12
4.3 Parsing numbers . . . 12
4.3.1 input-digits, input-ignore . . . 12
4.3.2 input-comparators . . . 13
4.3.3 input-[open|close]-uncertainty, input-uncertainty-signs . . . 13
4.3.4 input-uncertainty-divider . . . 13
4.3.5 parse-numbers . . . 13
4.3.6 retain-explicit-decimal-marker, retain-explicit-plus, retain-negative-zero, retain-zero-uncertainty . . . 13
4.4 Post-processing numbers . . . 14
4.4.1 exponent-mode, fixed-exponent . . . 14
4.4.2 exponent-thresholds . . . 14
4.4.3 drop-exponent, drop-uncertainty . . . 14
4.4.4 round-mode, round-precision, round-pad . . . 14

4.4.5 round-direction, round-half 15
4.4.6 uncertainty-round-direction 15
4.4.7 round-minimum 16
4.4.8 round-zero-positive 16
4.4.9 drop-zero-decimal 16
4.4.10 minimum-decimal-digits, minimum-integer-digits 16
4.4.11 Other Tests 16
4.5 Printing Numbers 17
4.5.1 group-digits, group-separator 17
4.5.2 group-minimum-digits 17
4.5.3 digit-group-size, digit-group-first-size, digit-group-other-size 17
4.5.4 output-decimal-marker 18
4.5.5 exponent-base, exponent-product 18
4.5.6 output-exponent-marker 18
4.5.7 uncertainty-[mode|separator], output-[open|close]-uncertainty 18
4.5.8 uncertainty-descriptors, uncertainty-descriptor-mode, uncertainty-descriptor-separator 18
4.5.9 simplify-uncertainty 19
4.5.10 bracket-ambiguous-numbers 19
4.5.11 negative-color 19
4.5.12 bracket-negative-numbers 19
4.5.13 tight-spacing 20
4.5.14 print-implicit-plus, print-[exponent|mantissa]-implicit-plus 20
4.5.15 print-unity-mantissa, print-zero-[exponent|integer] 21
4.5.16 zero-decimal-as-symbol, zero-symbol 21
4.6 Lists, products and ranges 21
4.6.1 list-[final|pair]-separator, list-separator 21
4.6.2 product-[mode|phrase|symbol] 21
4.6.3 range-open-phrase, range-phrase 21
4.6.4 [list|product|range]-exponents 22
4.6.5 [list|product|range]-units 22
4.6.6 [list|product|range]-[close|open]-bracket 22
4.6.7 [list|product|range]-independent-prefix 22
4.6.8 prefix-mode = combine-exponent 23
4.7 Complex Numbers 23
4.7.1 complex-mode 23
4.7.2 input-complex-roots 23
4.7.3 output-complex-root 23
4.7.4 complex-root-position 24
4.7.5 complex-angle-unit, complex-symbol-angle, complex-symbol-degree 24
4.7.6 print-complex-unity 24
4.7.7 Other tests 24
4.8 Angles 24
4.8.1 angle-mode 24

4.8.2 angle-separator . . . 24
4.8.3 fill-angle-[degrees|minutes|seconds] . . . 25
4.8.4 angle-symbol-[degree|minute|second] . . . 25
4.8.5 angle-symbol-over-decimal . . . 25
4.10 Using Units . . . 25
4.10.1 inter-unit-product . . . 25
4.10.2 per-mode, [display|inline]-per-mode, per-symbol, fraction-command, bracket-unit-denominator . . . 25
4.10.3 per-symbol-script-correction . . . 26
4.10.4 sticky-per . . . 26
4.10.5 qualifier-[mode|phrase] . . . 26
4.10.6 power-half-as-sqrt . . . 27
4.10.7 parse-units . . . 27
4.10.8 forbid-literal-units . . . 27
4.10.9 unit-font-command . . . 27
4.11 Quantities . . . 27
4.11.1 allow-quantity-breaks . . . 27
4.11.2 quantity-product . . . 27
4.11.3 prefix-mode, extract-mass-in-kilograms . . . 28
4.11.4 separate-uncertainty-units . . . 28
4.12 Tabular Material . . . 28
4.12.1 table-alignment-mode . . . 28
4.12.2 table-number-alignment . . . 28
4.12.3 table-format . . . 28
4.12.4 table-model-setup . . . 30
4.12.5 table-align-[comparator|exponent|uncertainty] . . . 30
4.12.6 table-align-text-[before|after] . . . 30
4.12.7 table-auto-round . . . 30
4.12.8 parse-numbers . . . 30
4.12.9 drop-exponent . . . 30
4.12.10 table-column-width, table-fixed-width . . . 32
4.12.11 table-text-alignment, table-alignment . . . 32
4.12.12 table-alignment . . . 32
4.13 Locale Options . . . 32
4.14 Preamble-only options . . . 32

10 Hints for using siunitx **33**
10.2 Adjusting \litre and \liter . . . 33
10.5 Expanding content in tables . . . 33
10.11 Special considerations for the \kWh unit . . . 33
10.12 Prefixes and small angles . . . 33
10.17 Demonstrating prefixes . . . 33

1 Introduction

This tries to follow the package documentation released 2025-07-09, with additional tests. Items remaining (generally ordered from hardest to easiest) are

- error: evaluate-expression, expression
- sansmath: package needed to evaluate two more tests
- The spacing in tables is not quite right. See the comment on &six_table_space
- font and spacing: several instances where we should make a font change or adjust the spacing, but don't.
- detect if loaded with `[=v2]` or `[=2021-04-09]` or earlier

2 siunitx for the impatient

12 345.678 90
0.3×10^{45}
$1 \pm 2\mathrm{i}$
$1.654 \times 2.34 \times 3.430$
$\mathrm{kg\,m\,s^{-1}}$
$\mathrm{kg\,m\,s^{-1}}$
$\mathrm{kg\,m/s}$
$\mathrm{kg\,m/(A\,s)}$
10, 20 and 30
0.13 mm, 0.67 mm and 0.80 mm
10 to 20
0.13 mm to 0.67 mm

Other tests

10, 20 and 30
$1.2^{\circ}3'4''$
Unsemantic: $\mathrm{m^2\,s}$
$\mu\mathrm{m}^2$
Semantic again: $0.094\,\pi\,\mathrm{mm\,mrad}$
$0.094\,\frac{1}{3}\,\mathrm{mm\,mrad}$
$0.094\,\pi/\mathrm{mm\,mrad}^3$

3 Using the siunitx package

3.1 Numbers

123
1234

$12\,345$
0.123
0.1234
$0.123\,45$
3.45×10^{-4}
-10^{10}
10, 30, 50 and 70
10×30
10 to 30

3.1.1 Other Tests

123×10^4
$123(3) \times 10^4$
$123(2)$
$123 \pm 2\mathrm{i}$
$123 + 234\mathrm{i}$
$(123(1) + 234(1)\mathrm{i}) \times 10^3$
$(234(1)\mathrm{i}) \times 10^3$
Pretty nonsensical stuff? $1.\pi e + 3$
1234.1234
3ξ
3ξ
3ξ
3ξ
3ξ
$1.23(1)$
$1.23(1)$
$1.23(\pi)$
$1.234(5)x\pi$
$1.234(5)x\pi$

3.1.2 Lists and ranges of numbers

$10\,\mathrm{m}$, $30\,\mathrm{m}$ and $45\,\mathrm{m}$
$10\,\mathrm{m} \times 30\,\mathrm{m} \times 45\,\mathrm{m}$
$10\,\mathrm{m}$ to $30\,\mathrm{m}$

3.2 Angles

10°
12.3°
4.5°
$1^\circ 2' 3''$
$1''$

10°
$-1'$

3.3 Units

3.3.1 unit

$\mathrm{kg\,m/s^2}$
$\mathrm{g_{polymer}\,mol_{cat}\,s^{-1}}$
$\mathrm{kg\,m\,s^{-2}}$
$\mathrm{g\,cm^{-3}}$
$\mathrm{V^2\,lm^3\,F^{-1}}$
$\mathrm{m^2\,Gy^{-1}\,lx^3}$
$\mathrm{H\,s}$

3.3.2 qty

$1.23\,\mathrm{J\,mol^{-1}\,K^{-1}}$
$0.23 \times 10^7\,\mathrm{cd}$
$1.99/\mathrm{kg}$
$1.345\,\frac{\mathrm{C}}{\mathrm{mol}}$

3.3.3 qtylist

$10\,\mathrm{m}$, $30\,\mathrm{m}$ and $45\,\mathrm{m}$

3.3.4 qtyproduct

$10\,\mathrm{m} \times 30\,\mathrm{m} \times 45\,\mathrm{m}$

3.3.5 qtyrange

$10\,\mathrm{m}$ to $30\,\mathrm{m}$

3.4 Complex numbers and quantities

$(123 + 234\mathrm{i}) \times 10^3$
$(123(1) + 234(1)\mathrm{i}) \times 10^3$
$3\mathrm{i}$
$(3\mathrm{i}) \times 10^4$

3.5 The unit macros

Table 1
Table 2
Table 3
Table 4

Table 1: SI base units

Unit	Macro	Symbol
ampere	`\ampere`	A
candela	`\candela`	cd
kelvin	`\kelvin`	K
kilogram	`\kilogram`	kg
metre	`\metre`	m
mole	`\mole`	mol
second	`\second`	s

Table 2: Coherent derived units

Unit	Macro	Symb	Unit	Macro	Symb
becquerel	`\becquerel`	Bq	newton	`\newton`	N
degreeCelsius	`\degreeCelsius`	°C	ohm	`\ohm`	Ω
coulomb	`\coulomb`	C	pascal	`\pascal`	Pa
farad	`\farad`	F	radian	`\radian`	rad
gray	`\gray`	Gy	siemens	`\siemens`	S
hertz	`\hertz`	Hz	sievert	`\sievert`	Sv
henry	`\henry`	H	steradian	`\steradian`	sr
joule	`\joule`	J	tesla	`\tesla`	T
katal	`\katal`	kat	volt	`\volt`	V
lumen	`\lumen`	lm	watt	`\watt`	W
lux	`\lux`	lx	weber	`\weber`	Wb

Table 3: Non-SI units

Unit	Macro	Symbol
astronomicalunit	`\astronomicalunit`	au
bel	`\bel`	B
dalton	`\dalton`	Da
day	`\day`	d
decibel	`\decibel`	dB
degree	`\degree`	°
electronvolt	`\electronvolt`	eV
hectare	`\hectare`	ha
hour	`\hour`	h
litre	`\litre`	L
liter	`\liter`	L
arcminute	`\arcminute`	′
minute	`\minute`	min
arcsecond	`\arcsecond`	″
neper	`\neper`	Np
tonne	`\tonne`	t

Table 4: SI prefixes

Prefix	Cmd	Symb	Power	Prefix	Cmd	Symb	Power
quecto	`\quecto`	q	10^{-30}	deca	`\deca`	da	10^{1}
ronto	`\ronto`	r	10^{-27}	hecto	`\hecto`	h	10^{2}
yocto	`\yocto`	y	10^{-24}	kilo	`\kilo`	k	10^{3}
zepto	`\zepto`	z	10^{-21}	mega	`\mega`	M	10^{6}
atto	`\atto`	a	10^{-18}	giga	`\giga`	G	10^{9}
femto	`\femto`	f	10^{-15}	tera	`\tera`	T	10^{12}
pico	`\pico`	p	10^{-12}	peta	`\peta`	P	10^{15}
nano	`\nano`	n	10^{-9}	exa	`\exa`	E	10^{18}
micro	`\micro`	µ	10^{-6}	zetta	`\zetta`	Z	10^{21}
milli	`\milli`	m	10^{-3}	yotta	`\yotta`	Y	10^{24}
centi	`\centi`	c	10^{-2}	ronna	`\ronna`	R	10^{27}
deci	`\deci`	d	10^{-1}	quetta	`\quetta`	Q	10^{30}

Bq^{2}
$\mathrm{J}^{2}\,\mathrm{lm}^{-1}$
$\mathrm{lx}^{3}\,\mathrm{V}\,\mathrm{T}^{3}$
H^{5}
$\mathrm{rad}^{4.5}$
$\mathrm{J\,mol}^{-1}\,\mathrm{K}^{-1}$
$\mathrm{J\,mol}^{-1}\,\mathrm{K}$
H^{-5}
Bq^{-2}
$\mathrm{kg}_{\mathrm{metal}}$
$0.1\,\mathrm{mmol(cat)\,kg(prod)}^{-1}$
$\frac{\cancel{\mathrm{kg}}\,\mathrm{m}}{\cancel{\mathrm{kg}}\,\mathrm{s}}$
$\mathrm{kg\,m\,s}^{-1}$
$\mathrm{kg\,m\,s}^{-1}$

3.6 Unit abbreviations

Table 5

Table 5: Abbreviated units

Unit	Abbreviation	Symbol
femtogram	`\fg`	fg
picogram	`\pg`	pg
nanogram	`\ng`	ng
microgram	`\ug`	µg
milligram	`\mg`	mg
gram	`\g`	g

Table 5: Abbreviated units (cont)

Unit	Abbreviation	Symbol
kilogram	`\kg`	kg
picometre	`\pm`	pm
nanometre	`\nm`	nm
micrometre	`\um`	µm
millimetre	`\mm`	mm
centimetre	`\cm`	cm
decimetre	`\dm`	dm
metre	`\m`	m
kilometre	`\km`	km
picometer	`\pm`	pm
nanometer	`\nm`	nm
micrometer	`\um`	µm
millimeter	`\mm`	mm
centimeter	`\cm`	cm
decimeter	`\dm`	dm
meter	`\m`	m
kilometer	`\km`	km
attosecond	`\as`	as
femtosecond	`\fs`	fs
picosecond	`\ps`	ps
nanosecond	`\ns`	ns
microsecond	`\us`	µs
millisecond	`\ms`	ms
second	`\s`	s
femtomole	`\fmol`	fmol
picomole	`\pmol`	pmol
nanomole	`\nmol`	nmol
micromole	`\umol`	µmol
millimole	`\mmol`	mmol
mole	`\mol`	mol
kilomole	`\kmol`	kmol
picoampere	`\pA`	pA
nanoampere	`\nA`	nA
microampere	`\uA`	µA
milliampere	`\mA`	mA
ampere	`\A`	A
kiloampere	`\kA`	kA
microlitre	`\ul`	µL
millilitre	`\ml`	mL
litre	`\l`	L

Table 5: Abbreviated units (cont)

Unit	Abbreviation	Symbol
hectolitre	`\hl`	hL
microliter	`\uL`	µL
milliliter	`\mL`	mL
liter	`\L`	L
hectoliter	`\hL`	hL
millihertz	`\mHz`	mHz
hertz	`\Hz`	Hz
kilohertz	`\kHz`	kHz
megahertz	`\MHz`	MHz
gigahertz	`\GHz`	GHz
terahertz	`\THz`	THz
millinewton	`\mN`	mN
newton	`\N`	N
kilonewton	`\kN`	kN
meganewton	`\MN`	MN
pascal	`\Pa`	Pa
kilopascal	`\kPa`	kPa
megapascal	`\MPa`	MPa
gigapascal	`\GPa`	GPa
milliohm	`\mohm`	mΩ
kiloohm	`\kohm`	kΩ
megaohm	`\Mohm`	MΩ
picovolt	`\pV`	pV
nanovolt	`\nV`	nV
microvolt	`\uV`	µV
millivolt	`\mV`	mV
volt	`\V`	V
kilovolt	`\kV`	kV
nanowatt	`\nW`	nW
microwatt	`\uW`	µW
milliwatt	`\mW`	mW
watt	`\W`	W
kilowatt	`\kW`	kW
megawatt	`\MW`	MW
gigawatt	`\GW`	GW
microjoule	`\uJ`	µJ
millijoule	`\mJ`	mJ
joule	`\J`	J

Table 5: Abbreviated units (cont)

Unit	Abbreviation	Symbol
kilojoule	`\kJ`	kJ
millielectronvolt	`\meV`	meV
electronvolt	`\eV`	eV
kiloelectronvolt	`\keV`	keV
megaelectronvolt	`\MeV`	MeV
gigaelectronvolt	`\GeV`	GeV
teraelectronvolt	`\TeV`	TeV
kilowatt hour	`\kWh`	kW h
femtofarad	`\fF`	fF
picofarad	`\pF`	pF
nanofarad	`\nF`	nF
microfarad	`\uF`	µF
millifarad	`\mF`	mF
farad	`\F`	F
femtohenry	`\fH`	fH
picohenry	`\pH`	pH
nanohenry	`\nH`	nH
microhenry	`\uH`	µH
millihenry	`\mH`	mH
henry	`\H`	H
nanocoulomb	`\nC`	nC
microcoulomb	`\uC`	µC
millicoulomb	`\mC`	mC
coulomb	`\C`	C
microtesla	`\uT`	µT
millitesla	`\mT`	mT
tesla	`\T`	T
kelvin	`\K`	K
decibel	`\dB`	dB

Table 6

Table 6: Binary prefixes

Prefix	Command	Symbol	Power
kibi	`\kibi`	Ki	10
mebi	`\mebi`	Mi	20
gibi	`\gibi`	Gi	30
tebi	`\tebi`	Ti	40
pebi	`\pebi`	Pi	50
exbi	`\exbi`	Ei	60
zebi	`\zebi`	Zi	70
yobi	`\yobi`	Yi	80

Table 7: Standard behaviour of the `S` column type.

Some Values
2.3456
34.2345
−6.7835
90.473
5642.5
1.2×10^{3}
10^{4}

3.7 Creating new macros

67 890°
67 890°
kg^{4}
m^{4}
$1.234\,\mathrm{g_{pol}\,mol_{cat}^{-1}\,h^{-1}}$

3.7.1 Other Tests

kg kg km kg
a
a
a
e
e
a
a
km
$\mathrm{kg\,m\,s^{-1}}$
$\mathrm{kg\,m\,s^{-1}}$
$\mathrm{kg\,m\,s^{-1}}$
$\mathrm{kg\,m\,s^{-1}}$
$\mathrm{kg\,m\,s^{-1}}$
$\mathrm{kg\,m\,s^{-1}}$
$\mathrm{kg\,m\,s^{-1}}$
$\mathrm{kg\,m\,s^{-1}}$
$\mathrm{kg\,m\,s^{-1}}$
$\mathrm{kg\,m\,s^{-1}}$

3.8 Tabular material

Table 7
Table 8

Table 8: Detection of surrounding material in an `S` column.

Some Values
12.34
975.31
44.268[a]

Table 9: Text before and after numbers.

Values
2.3456
34.2345[a]
34.2345[a]
56.7835
now 90.473

Table 9
Table 10

4 Package control options

4.2 Printing

4.2.1 mode, number-mode, unit-mode

4.2.2 reset-text-[family|series|shape]

1234
1234
1234
1234
1234
1234

Table 10: Controlling complex alignment with the tablenum macro.

Heading	Heading
Info	More info
Info	More info
12.34	
333.5567	
4563.21	

Heading	Heading
88.999	aaa bbb
33.435	ccc ddd

4.2.3 propagate-math-font, reset-math-version

kg
kg
kg
kg

4.2.4 text-[family|series]-to-math

kg
kg
kg
kg

4.2.5 text-font-command

123 456 789 kV cm^{-1}
123 456 789 kV cm^{-1}

4.2.6 text-[sub|super]script-command

kg$_{\text{polymer}}$
kg$_{\text{polymer}}$

4.2.7 color, number-color, unit-color

Some text
4 kg
More text
4 kg
4 kg
4 kg
Still red here!
1 g
1 g
1 g
1 g
$-1^{\circ}2'3''$
1, 2 and 3
1 by 2
1 to 2
$1.234(5) \times 6.78(9)$

4.3 Parsing numbers

4.3.1 input-digits, input-ignore

1245

4.3.2 input-comparators

<10
$\gg 5\,\mathrm{m}$
≤ 0.12

4.3.3 input-[open|close]-uncertainty, input-uncertainty-signs

9.99(9)
9.99(9)
9.99(9)
123.0(45)
12.3(60)
123.4(12)
123.4(12)
$123.4 \pm 1.2 \pm 4.5$
$123.4 \pm 1.2 \pm 4.5$

4.3.4 input-uncertainty-divider

$10.56^{+0.12}_{-0.34}$
$123.4^{+4.6}_{-7.8}$
$10.56^{+0.01}_{-0.02} \pm 0.03$
$6.45 \pm 0.02^{+0.03}_{-0.04}$
Also:
$5.6^{+1.2}_{-3.4}$
$78.56^{+0.12}_{-0.34}$
$123.4 \pm 1.2 \pm 4.5 \pm 6.7$
$123.4 \pm 1.2 \pm 4.5 \pm 6.7$

4.3.5 parse-numbers

$\sqrt{2}$
$\sqrt{3}\,\mathrm{m}$

Other tests

π
2π
$\pi/3$

4.3.6 retain-explicit-decimal-marker, retain-explicit-plus, retain-negative-zero, retain-zero-uncertainty

345
+345
0

-0
12.3
12.3(0)
456
0.789
456
0.789

4.4 Post-processing numbers

4.4.1 exponent-mode, fixed-exponent

0.001
0.0100
1200
1×10^{-3}
1.00×10^{-2}
1.200×10^{3}
1×10^{-3}
10.0×10^{-3}
1.200×10^{3}
$0.000\,01 \times 10^{2}$
$0.000\,100 \times 10^{2}$
12.00×10^{2}
1.23×10^{4}
12 300
$(0.000\,500 + 1.000\mathrm{i}) \times 10^{3}$

4.4.2 exponent-thresholds

This is unique to version 3.

4.4.3 drop-exponent, drop-uncertainty

0.01(2)
0.01
0.01×10^{3}
0.01

4.4.4 round-mode, round-precision, round-pad

1.234 56
14.23
0.123 45(9)
0.001 234 5
places:
1.235

14.230
0.123 45(9)
0.001
figures:
1.23
14.2
0.123 45(9)
10 000
0.001 23
uncertainty:
0.123 45(9)
0.1235(2)
0.123(2)
12 300(100)
123.46(1)
1230(10)
1235(1)
10(1)
100(10)
figures:
12.30
12.34
12.3
Also
$0.12 \times 0.46 \times 0.79$

4.4.5 round-direction, round-half

0.05
0.05
0.05
0.04
0.06
0.05
0.06
0.05
0.06
0.04

4.4.6 uncertainty-round-direction

0.123(41)
0.123(41)
0.12(5)

4.4.7 round-minimum

0.01
0.00
0.00
0.01
<0.01
>−0.01
<0.01

4.4.8 round-zero-positive

0.00
−0.00

4.4.9 drop-zero-decimal

2.0
2.1
2
2.1

4.4.10 minimum-decimal-digits, minimum-integer-digits

123
123
0123
0.123
0.123
0.1230

4.4.11 Other Tests

1
456
0.789
456
.789
Also:
1.00
1.0
1.00
-52.0 ± 0.4

4.5 Printing Numbers

4.5.1 group-digits, group-separator

12 345.678 90
12345.67890
12345.678 90
12 345.67890
12 345
12,345
12 345
12 345

4.5.2 group-minimum-digits

1234
12 345
1234
12 345
1234.5678
12 345.678 90
1234.5678
12 345.678 90

Other tests

1234
1 234
1234.5678
1 234.567 8

4.5.3 digit-group-size, digit-group-first-size, digit-group-other-size

1 234 567 890
12345 67890
1 23 45 67 890

Other tests

1 234 567 890.123 456 789 0
12 3456 7890.1234 5678 90
1 23 45 67890
12345 67890
123 45678 90
12345 67890

4.5.4 output-decimal-marker

1.23
1,23

4.5.5 exponent-base, exponent-product

1×10^2
$1 \cdot 10^2$
1×2^2

4.5.6 output-exponent-marker

$1e2$
1e2
1E2
1E2

4.5.7 uncertainty-[mode|separator], output-[open|close]-uncertainty

123.45(120)
0.035(14)
123.45(1.20)
0.035(0.014)
123.45(1.20)
0.035(14)
1.234 [5]
1.234 [5]
1.234 ± 0.005
1.234 ± 0.005
8.2 [1.3]
8.2 [1.3]
8.2 ± 1.3
8.2 ± 1.3
1.200 [1]
1.200 ± 0.001
$\boxed{\begin{array}{l} 67\,890 \ \pm \\ 12\,345 \\ 67\,890 \pm 12 \end{array}}$ 345

4.5.8 uncertainty-descriptors, uncertainty-descriptor-mode, uncertainty-descriptor-separator

$1.2 \pm 0.3 \pm 0.4$
$1.2 \pm 0.3\ (\mathrm{sys}) \pm 0.4\ (\mathrm{stat})$
$1.2 \pm 0.3_{\mathrm{sys}} \pm 0.4_{\mathrm{stat}}$
$1.2 \pm 0.3(\mathrm{sys}) \pm 0.4(\mathrm{stat})$

1.2 ± 0.3 sys ± 0.4 stat
1.2 ± 0.3
1.2 ± 0.3 (sys) ± 0.4 (stat) ± 0.5
1.2 ± 0.3
1.2 ± 0.3 (sys) ± 0.4 (stat)

4.5.9 simplify-uncertainty

$10.56^{+0.02}_{-0.02}$
10.56 ± 0.02

4.5.10 bracket-ambiguous-numbers

1×10^{10}
$(2\mathrm{i}) \times 10^{10}$
$(1 + 2\mathrm{i}) \times 10^{10}$
$1 + 2\mathrm{i} \times 10^{10}$
$(1.2 \pm 0.3) \times 10^{4}$
$1.2 \pm 0.3 \times 10^{4}$
$1.234 \pm 0.005 \times 10^{-4}$

4.5.11 negative-color

$-15\,673$
$-15\,673$
also:
0
-0
-10
-10
1
-1
$3\mathrm{i}$
$-3\mathrm{i}$
$1 + 3\mathrm{i}$
$-1 + 3\mathrm{i}$
$1 - 3\mathrm{i}$
$-1 - 3\mathrm{i}$

4.5.12 bracket-negative-numbers

$-15\,673$
$(15\,673)$
$-10\,\mathrm{m}$
$(10)\,\mathrm{m}$
$\ll(5)$
$(13(1))$

(13(1))
(10^{10})
(2×10^{10})
1
(1)
3i
((3i))
1 + 3i
(1) + 3i
(1(3i))
((1)(3i))

4.5.13 tight-spacing

2×10^3
$2{\times}10^3$

4.5.14 print-implicit-plus, print-[exponent|mantissa]-implicit-plus

a
345
+345
+345
345
b
1×10^2
$+1 \times 10^{+2}$
$+1 \times 10^2$
$1 \times 10^{+2}$
c
1×10^2
$+1 \times 10^{+2}$
$+1 \times 10^2$
$1 \times 10^{+2}$
d
-1×10^2
$-1 \times 10^{+2}$
-1×10^2
$-1 \times 10^{+2}$
e
1×10^{-2}
$+1 \times 10^{-2}$
$+1 \times 10^{-2}$
1×10^{-2}

4.5.15 print-unity-mantissa, print-zero-[exponent|integer]

1×10^4
10^4
444
444×10^0
0.123
.123
10^4
10^4
1
1
1×10^0
10^0

4.5.16 zero-decimal-as-symbol, zero-symbol

123.00
123.—
123—
123.[—]

4.6 Lists, products and ranges

4.6.1 list-[final|pair]-separator, list-separator

0.1, 0.2 and 0.3
0.1; 0.2 and 0.3
0.1, 0.2, 0.3
0.1 and 0.2 and finally 0.3
0.1 and 0.2
0.1, and 0.2
0.1, 0.2 and 0.3

4.6.2 product-[mode|phrase|symbol]

$5 \times 100 \times 2$
$5 \cdot 100 \cdot 2$
5 by 100 by 2
5 BY 100 BY 2

4.6.3 range-open-phrase, range-phrase

5 to 100
5–100
10 to 12
from 5 to 100
5–100

4.6.4 [list|product|range]-exponents

5×10^3, 7×10^3, 9×10^3 and 1×10^4
$5 \times 10^3 \times 7 \times 10^3 \times 9 \times 10^3 \times 1 \times 10^4$
5×10^3 to 7×10^3
(5, 7, 9 and 10) $\times 10^3$
$(5 \times 7 \times 9 \times 10) \times 10^3$
(5 to 7) $\times 10^3$
from (5 to 7) $\times 10^3$
5, 7, 9 and 10×10^3
$5 \times 7 \times 9 \times 10 \times 10^3$
5 to 7×10^3

4.6.5 [list|product|range]-units

2 T, 4 T, 6 T and 8 T
(2, 4, 6 and 8) T
2 T, 4 T, 6 T and 8 T
2, 4, 6 and 8 T
2 °C to 4 °C
(2 to 4) °C
2 °C to 4 °C
2 to 4 °C
2 m × 3 m × 4 m
(2 × 3 × 4) m
2 m × 3 m × 4 m
2 × 3 × 4 m
(2 × 3 × 4) m^3
2 × 3 × 4 m^3

4.6.6 [list|product|range]-[close|open]-bracket

[5×10^3, 7×10^3, 9×10^3 and 1×10^4] s
{$5 \times 10^3 \times 7 \times 10^3 \times 9 \times 10^3 \times 1 \times 10^4$} m
⟨2 to 4⟩ °C
5×10^3, 7×10^3, 9×10^3 and 1×10^4
$5 \times 10^3 \times 7 \times 10^3 \times 9 \times 10^3 \times 1 \times 10^4$
(2 to 4) °C

4.6.7 [list|product|range]-independent-prefix

a
1×10^3 W to 1×10^9 W
1×10^3 W and 1×10^9 W
1×10^3 W × 1×10^9 W
b
1×10^3 W to 1×10^9 W

1×10^3 W and 1×10^9 W
1×10^3 W $\times$ 1×10^9 W
c
1000 W to 1 000 000 000 W
1 kW to 1 000 000 kW
1000 W and 1 000 000 000 W
1 kW and 1 000 000 kW
1000 W $\times$ 1 000 000 000 W
1 kW $\times$ 1 000 000 kW
d
1 kW to 1 GW
1 kW and 1 GW
1 kW $\times$ 1 GW

4.6.8 prefix-mode = combine-exponent

1700 g
1.7×10^3 g
1700 g
1.7 kg
1.700×10^3 g
1.7×10^3 g

4.7 Complex Numbers

4.7.1 complex-mode

$1 + \mathrm{i}$
$1\angle 45°$
$1 + \mathrm{i}$
$0.71 + 0.71\mathrm{i}$
$1.414\,213\,562\,373\,095\angle 45°$
$1\angle 45°$

4.7.2 input-complex-roots

$9.99 + 88.8\mathrm{i}$
$9.99 + 88.8\mathrm{i}$

4.7.3 output-complex-root

$1 + 2\mathrm{i}$
$1 + 2i$
$1 + 2i$
$1 + 2j$

4.7.4 complex-root-position

$67 - 0.9\mathrm{i}$
$67 - \mathrm{i}0.9$
$67 - 0.9\mathrm{i}$

4.7.5 complex-angle-unit, complex-symbol-angle, complex-symbol-degree

$1\angle 1^\circ\,\Omega$
$1\angle 1\,\Omega$
$1\angle 1\,\mathrm{d}\,\Omega$

4.7.6 print-complex-unity

$\mathrm{i}\,\Omega$
$1\mathrm{i}\,\Omega$
$1\mathrm{i}\,\Omega$

4.7.7 Other tests

Real numbers as complex:
1
1
-1
1
-1
1
$-1 - 0.000\,000\,000\,000\,206\,761\,537\,356\,616\,7\mathrm{i}$
$1 + 0.000\,000\,000\,000\,413\,523\,074\,713\,233\,4\mathrm{i}$
$1\angle 1.570\,796\,326\,794\,897$

4.8 Angles

4.8.1 angle-mode

2.67°
$2^\circ 3' 4''$
$2^\circ 40' 12''$
$2^\circ 3' 4''$
2.67°
$2.051\,111\,111\,111\,111^\circ$

4.8.2 angle-separator

$6^\circ 7' 6.5''$
$6^\circ\ 7'\ 6.5''$

4.8.3 fill-angle-[degrees|minutes|seconds]

-1°
$-2'$
$-3''$
-1°
$-0^{\circ}2'$
$-0^{\circ}3''$
$-1^{\circ}0'$
$-2'$
$-0'3''$
$-1^{\circ}0''$
$-2'0''$
$-3''$
45.697°
45.697°

4.8.4 angle-symbol-[degree|minute|second]

$6^{\circ}7'6.5''$
$6\,\mathrm{d}7\,\mathrm{m}6.5\,\mathrm{s}$

4.8.5 angle-symbol-over-decimal

45.697°
$6^{\circ}7'6.5''$
$45\overset{\circ}{.}697$
$6^{\circ}7'6\overset{''}{.}5$
$6^{\circ}7'6\overset{''}{.}5$

4.10 Using Units

4.10.1 inter-unit-product

$\mathrm{F}^2\,\mathrm{lm}\,\mathrm{cd}$
$\mathrm{F}^2 \cdot \mathrm{lm} \cdot \mathrm{cd}$
$\mathrm{F}^2 \cdot \mathrm{lm} \cdot \mathrm{cd}$

4.10.2 per-mode, [display|inline]-per-mode, per-symbol, fraction-command, bracket-unit-denominator

$\mathrm{J}\,\mathrm{mol}^{-1}\,\mathrm{K}^{-1}$
$\mathrm{m}\,\mathrm{s}^{-2}$
$\frac{\mathrm{J}}{\mathrm{mol}\,\mathrm{K}}$
$\frac{\mathrm{J}}{\mathrm{mol}\,\mathrm{K}}$
$\frac{\mathrm{m}}{\mathrm{s}^2}$

$\mathrm{A\,mol^{-1}\,s}$
$\mathrm{A\,s\,mol^{-1}}$
$\mathrm{J/(mol\,K)}$
$\mathrm{m/s^2}$
J div (mol K)
$\mathrm{J/mol\,K}$
$\mathrm{J/mol/K}$
$\mathrm{10\,m^{-1}}$
$\mathrm{20\,m/s}$
$\mathrm{30\,J\,mol^{-1}\,K^{-1}}$
$\mathrm{J/(mol\,K)}$

$$\frac{\mathrm{J}}{\mathrm{mol\,K}}$$

$\mathrm{J/(mol\,K)}$
$\frac{\mathrm{J}}{\mathrm{mol\,K}}$

$$\mathrm{J/(mol\,K)}$$

$$\mathrm{J/(mol\,K)}$$

Other tests:
$\mathrm{m\,L/s}$
J over mol K

4.10.3 per-symbol-script-correction

$\mathrm{cm^3/g}$
$\mathrm{cm^3/g}$

4.10.4 sticky-per

$\mathrm{Pa\,Gy^{-1}\,H}$
$\mathrm{Pa\,Gy^{-1}\,H^{-1}}$

4.10.5 qualifier-[mode|phrase]

$\mathrm{kg^2_{pol}\,mol^{-1}_{cat}\,h^{-1}}$
$\mathrm{kg(pol)^2\,mol(cat)^{-1}\,h^{-1}}$
$\mathrm{kg^2_{pol}\,mol^{-1}_{cat}\,h^{-1}}$
kg $\mathrm{pol^2}$ mol $\mathrm{cat^{-1}\,h^{-1}}$
dBi
$\mathrm{kg^2_{pol}\,mol^{-1}_{cat}\,h^{-1}}$
$\mathrm{kg(pol)^2\,mol(cat)^{-1}\,h^{-1}}$
dBi
kg $\mathrm{pol^2}$ mol $\mathrm{cat^{-1}\,h^{-1}}$
kg of $\mathrm{pol^2}$ mol of $\mathrm{cat^{-1}\,h^{-1}}$ $\mathrm{kg}\mathit{by}\mathrm{pol^2\,mol}\mathit{by}\mathrm{cat^{-1}\,h^{-1}}$
$\mathrm{kg(pol)^2}$
$\mathrm{kg(pol)^2}$

$\mathrm{kg\ pol}^2$
$\mathrm{kg\ pol}^2$
kgpol^2
kgpol^2

4.10.6 power-half-as-sqrt

$\mathrm{Hz}^{0.5}$
$\sqrt{\mathrm{Hz}}$

4.10.7 parse-units

$300\,\mathrm{MHz}$
$300\,\mathrm{MHz}$
$\mathrm{m\,s}^{-1}$
$\mathrm{m/s}$

4.10.8 forbid-literal-units

$\mathrm{m/s}$
$\mathrm{m\,s}^{-1}$

4.10.9 unit-font-command

lm
lm

4.11 Quantities

4.11.1 allow-quantity-breaks

X
$10\,\mathrm{m}$
X
$10\,\mathrm{m}$
X
$10\,\mathrm{m}$
X 10
m

4.11.2 quantity-product

$2.67\,\mathrm{F}$
$2.67\ \mathrm{F}$
$2.67\mathrm{F}$
$2.67\,\mathrm{F}$
$2.67\ \mathrm{F}$
$2.67\times\mathrm{F}$

2.67×F
also
(3.7 ± 4.5)×F

4.11.3 prefix-mode, extract-mass-in-kilograms

$1 \times 10^{3}\,\mathrm{m\,s}$
$1\,\mathrm{km\,s}$
$10\,\mathrm{kg}^{2}\,\mathrm{ds}$
$10 \times 10^{-1}\,\mathrm{kg}^{2}\,\mathrm{s}$
$7.5 \times 10^{-3}\,\mathrm{kg}$
$10\,\mathrm{kg}^{2}\,\mathrm{ds}$
$10 \times 10^{5}\,\mathrm{g}^{2}\,\mathrm{s}$
$7.5\,\mathrm{g}$
$\mathrm{mL\,mol}^{-1}\,\mathrm{dA}$
$\mathrm{mL\,mol}^{-1}\,\mathrm{dA}$
$\mathrm{kg}^{2}\,\mathrm{ds}$
$\mathrm{Mg}^{2}\,\mathrm{ds}$
$\mathrm{Mg}^{2}\,\mathrm{ds}$
$\mu\mathrm{g}^{2}\,\mathrm{ds}$
$\mu\mathrm{g}^{2}\,\mathrm{ds}$
$\mathrm{Mg}^{-2}\,\mathrm{ds}$
$\mathrm{Mg}^{-2}\,\mathrm{ds}$
$\mu\mathrm{g}^{-2}\,\mathrm{ds}$
$\mu\mathrm{g}^{-2}\,\mathrm{ds}$

4.11.4 separate-uncertainty-units

$(12.3 \pm 0.4)\,\mathrm{kg}$
$(12.3 \pm 0.4)\,\mathrm{kg}$
$12.3\,\mathrm{kg} \pm 0.4\,\mathrm{kg}$
$12.3 \pm 0.4\,\mathrm{kg}$

4.12 Tabular Material

4.12.1 table-alignment-mode

Table 12

4.12.2 table-number-alignment

Table 13

4.12.3 table-format

Table 14

Table 12: Parsing without aligning in an `S` column.

Decimal-centered	Simple centering
12.345	12.345
6.78	6.78
−88.8(9)	−88.8(9)
4.5×10^3	4.5×10^3

Table 13: Aligning the `S` column.

Some Values	Some Values	Some Values	Some Values
2.3456	2.3456	2.3456	2.3456
34.2345	34.2345	34.2345	34.2345
56.7835	56.7835	56.7835	56.7835
90.473	90.473	90.473	90.473

Table 14: Reserving space in `S` columns.

Values	Values	Values	Values	Values	Values
2.3	2.3	2.3(5)	2.3 ± 0.5	2.3	2.3×10^8
34.23	34.23	34.23(4)	34.23 ± 0.04	34.23	34.23
56.78	56.78	56.78(3)	56.78 ± 0.03	−56.78	56.78×10^3
3.76	3.76	3.76(2)	3.76 ± 0.02	±3.76	10^6

Table 15: The `table-align-exponent` option

Header	Header
1.2×10^{3}	1.2×10^{3}
1.234×10^{56}	1.234×10^{56}

Table 16: The `table-align-uncertainty` option.

Header	Header
1.2 ± 0.1	1.2 ± 0.3
1.234 ± 0.005	1.234 ± 0.005

4.12.4 table-model-setup

4.12.5 table-align-[comparator|exponent|uncertainty]

Table 15
Table 16
Table 17
Table 18

4.12.6 table-align-text-[before|after]

Table 19

4.12.7 table-auto-round

Table 20

4.12.8 parse-numbers

Table 21

4.12.9 drop-exponent

Table 22

Table 17: The `table-align-comparator` option.

Header	Header
> 1.2	>1.2
<12.34	<12.34

Table 18: Reserving space for comparators in `S` columns.

Values	Values	Values
2.3	$< 2.3 \times 10^8$	$<2.3 \times 10^8$
34.23	$= 34.23$	$=34.23$
56.78	$\geq 56.78 \times 10^3$	$\geq 56.78 \times 10^3$
3.76	$\gg 10^6$	$\gg 10^6$

Table 19: Closing notes up to text.

Values	Values
2.3456	2.3456
[a] 4.234	[a] 4.234
[b] 0.78	[b] 0.78
[d] 88	[d] 88

Values	Values
2.3456	2.3456
34.234 [a]	34.234 [a]
56.78 [b]	56.78 [b]
90.4 [c]	90.4 [c]
88 [d]	88 [d]

Table 20: The `table-auto-round` option.

Header	Header
1.2	1.200
1.2345	1.235

Table 21: Aligning without parsing.

Some values	Some values	Some values	Some values
2.35	2.35	2.35	2.35
34.234	34.234	34.234	34.234
56.783	56.783	56.783	56.783
3.762	3.762	3.762	3.762
$\sqrt{2}$	$\sqrt{2}$	$\sqrt{2}$	$\sqrt{2}$

Table 22: The `drop-exponent` option.

Header	Header / 10^3
1.2×10^3	1.2
3×10^2	0.3
1.0×10^4	10

Table 23: Fixed-width columns.

Flexible	Fixed
1.23	1.23
45.6	45.6

Table 24: Right-aligning under a heading.

Long header
12.33
2
1234

4.12.10 table-column-width, table-fixed-width

Table 23
Table 24

4.12.11 table-text-alignment, table-alignment

Table 25

4.12.12 table-alignment

4.13 Locale Options

1.234 m
6,789 m

4.14 Preamble-only options

These are in a different file. We check that trying to change the options has no effect here:
1, 2 and 3
$1 \times 2 \times 3$

Table 25: Aligning text in `S` columns.

Values	Values	Values
992.435	992.435	992.435
7734.2344	7734.2344	7734.2344
56.7834	56.7834	56.7834
3.7462	3.7462	3.7462

Table 26: Values as macros in `S` columns.

Some Values
12 348.812 34
12 348.812 34
12348.81234
1234 8.8 1234

Table check:

Decimal-centered	Simple centering
12.345	12.345
6.78	6.78
−88.8(9)	−88.8(9)
4.5×10^3	4.5×10^3

10 Hints for using siunitx

10.2 Adjusting \litre and \liter

SUCCESS

10.5 Expanding content in tables

Table 26

10.11 Special considerations for the \kWh unit

kWh
$\mathrm{kW\,h\,m^{-1}}$
$\mathrm{kWh\,m^{-1}}$
$\mathrm{kWh\,m^{-1}}$
$\mathrm{cd\,kW^{-1}\,h}$
$\mathrm{cd\,kW^{-1}\,h^{-1}}$
$\mathrm{cd\,kW^{-1}\,h^{-1}}$

10.12 Prefixes and small angles

$1 \times 10^{-3}\,\mathrm{as}$
$1\,\mathrm{mas}$

10.17 Demonstrating prefixes

Y
10^{24}
ab